

xMCF

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues [#95](#) / [#104](#)
2026-09-16

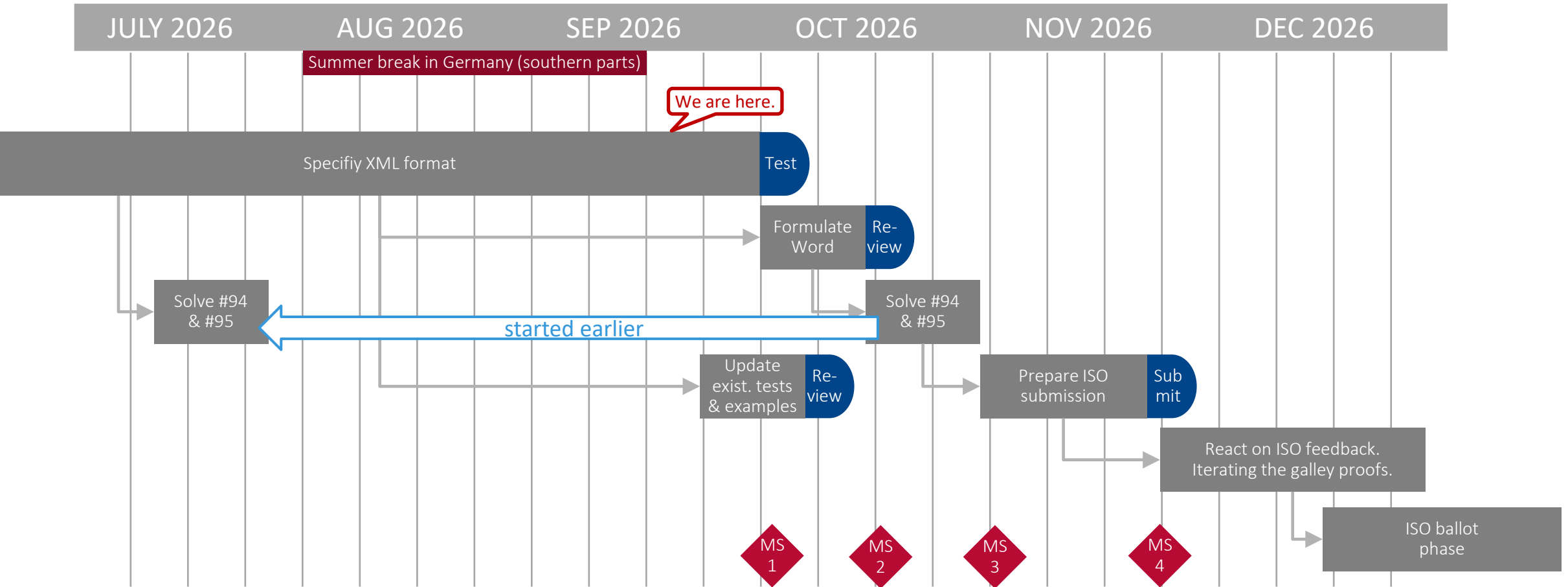
Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-09-16_VideoConference/README.md

- Welcome & Flying over today's topics
- Timeline for next release
- Work in progress — focus on issues [#95](#) / [#94](#) / [#106](#) / [#105](#) / [#61](#), as discussed last time.
- Focus on XML / DOC suggestions, XSD- / Schematron-definition & examples
- Steps we should take next



Timeline for the next release



Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.

Issue #95: Add persistent identifier to `connection_group`

as of 2026-07-01
≙ 2026-07-08

- Original request: „Up to now, `connection_groups` have an integer identifier which is local to each `χMCF` file, and not persistent.
Under PDM considerations, an optional persistent alphanumerical id would help to increase the process reliability ...”
- This use case has been very well justified.
- However, there are other use cases of great practical relevance where the introduction of a new mandatory attribute requiring attention would result in a significant additional workload, without this being offset by any benefit.
- The group decided to **continue the discussion** at the **next meeting** with the question:
‘What if the attribute (persistent identifier) were just *optional*?’ → See next slide.
- The following important point was also raised during the discussion:
<connected_to/> lists the connected parts in *no particular order*.
The order may change even after a simple im- / export sequence.
If the stacking sequence is important, always use the <stacking/> element!
→ 2026-09-06: An according [issue #112](#) was created.



Issue #95: Add persistent identifier to `connection_group`

(continued) ‘What if the attribute (persistent identifier) were just *optional*?’

Results of
2026-07-22
& 2026-07-28

- PDM Use Cases: Exchange with Partner and recognize a known connection group upon re-import.
- CAD Use Cases: Creating, Deleting, Splitting, Merging, Editing of connections, incl. changing the connected parts.
- PEP is “owner” of the unique connection group IDs, just as with the part codes.
(Single Point of Truth to guarantee uniqueness!)
PDM system, as “incarnation” of the PEP, is the tool which creates and keeps authority on such IDs.
- Difference between part codes and connection group IDs:
For CAD data, there is (usually) only *one* CAD authoring system, which is tightly integrated with PDM.
For connection data, this frequently is different:
Many tools are established for manipulating connections, which are not tightly PDM-integrated.
- Solution ideas:
 - a. PDM system offers and tool attaches to a standardized interface for querying suitable connection group ID.
 - b. Tools could exchange one xMCF file per connection group
(which renders the connection group ID as meaningless, as it is currently.)
 - c. More ideas? --- no → **Concept is confirmed (2026-09-16).** → **Text for spec update to be drafted in comments to issue #95.**
- Integer connection group *id* shall be supplemented with an alphanumeric *label*.
2026-09-09: ⇒ XSD to be updated. / **2026-09-16: SH added according change to GitHub, this morning.**



Issue #94: Add instance identifier to a connector

- Original request: „In order to keep consistency between the PDM system and the connection data, it is required to have an instance id for the rivets“ (Rivets or any other connection method in which a separate physical component establishes the connection.)
- The standard documents says in clause “8.2.2 Attribute `label`”: “Any connection should have an attribute called `label`, which identifies it throughout the entire CAx process, maybe even throughout the complete product lifecycle, including manufacturing. It is not necessary that these labels are unique.”

Remark 1: The use cases in which “`label`” is not unique are quite specific and rare.

Remark 2: “`label`” must not be confused with “`part_code`”, which frequently is *not* unique. Example:

```
<connection_0d label="BOLT_135" part_code="BOLT-NUT-Combi #17"> ... <!-- only label is unique -->
  <threaded_connection ... part_code="M10x50 8.8"/> <!-- not unique -->
</connection_0d>
<connection_0d label="BOLT_136" part_code="BOLT-NUT-Combi #17"> ... <!-- only label is unique -->
  <threaded_connection ... part_code="M10x50 8.8"/> <!-- not unique -->
</connection_0d>
```

In PDM, e.g. `label="BOLT_135"` would denote an *instance* combination of e.g. a bolt, some washers and a nut.

However, a part code *for this combination* is needed and not yet supported by xMCF.

⇒ Suitable place for the `part_code` for the combination to be found in xMCF syntax – at both, the actual connection plus its template, if there is one.

Suggestion: Put the `part_code` as shown above in blue, at both positions.

— **No**, it should be an attribute of `<threaded_connection/>`, see next slide!

2026-09-09: For discussion & validation, see following slides!

#94 wrt. templates

- The combination's `part_code` should *not* reside on `<connection_0d/>`, because there it would also be accessible for e.g. spotwelds. Instead, it should be an attribute of `<threaded_connection/>`. Thus, another name is needed for the combination's `part_code`, namely "`combination_part_code`".
- In case a connection template is used, the `part_code` of a connection may be given there.
- The minimal content of a `<connection_0d/>` would then be:
 - Label (highly recommended)
 - Template (as mentioned above)
 - `<loc/>`
- The part code of the connection (e.g. the bolt) and its components (screw, nut, washers) are then given only once, i. e. in the template definition.
- Remark: In any case, if there are different components of connections with the same combination's part code, this is an inconsistency and must be prevented by the exporting system.
⇒ Schematron rule to be added.
- PDM Systems would differentiate e.g. Screw-Nut-Combinations according to the distance between screw head and nut. Since xMCF does not cover this kind of distance, as more general geometric parameters too, tools cannot detect this situation.
- To-Dos: Extend XSD. Extend the Schema. Provide example file(s): With & without template. Formulate a text fragment for the spec.

Add this to the χ MCF Specification

- Label is optional, but highly recommended, especially in cases, when different tools or partners are involved.

2026-09-15: An according issue #113 was created.

9.1.1.1 General

The possible XML attributes of the `<connection_0d/>` element, which describes a point connection (0D connection), are specified in Table 32:

Table 32 — Attributes of element `<connection_0d/>`

Attributes	Type	Use	Constraint
label	Alphanumeric	Optional	-
ident	Integer	Optional	positive, unique within a χ MCF file
<u>quality_control</u>	Alphanumeric	Optional	See clause 8.4.

9.1.1.2 Attribute “label”

The label defines the human readable identification of a connection. It can contain a description of the connection or simply an index as an integer.

Other (ISO) Standards

[#61](#) suggests to analyze existing standards in this context, such as

- [ISO 13584-511:2006 — Industrial automation systems and integration — Parts library — Part 511: Mechanical systems and components for general use — Reference dictionary for fasteners](#)
— a copy was provided to the WG — for internal use, only.
- [ISO 1891:2009 — Fasteners — Terminology](#)
— a copy was provided to the WG — for internal use, only.

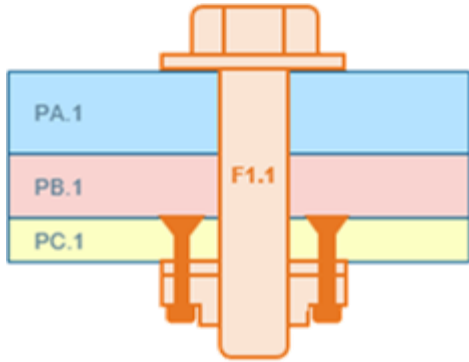
A sincere *thank you* to all donors!

- 2026-09-15: An according [issue #114](#) was created for this analysis.
- Do we need / want to consider more standards?

Issue #106: Support complex rivet-nut fasteners

Due to the composition of today's attendance list, the focus was placed on issue #106.

The task is to describe a design like this:



GH outlines a solution and invites discussion:

1. The nut is riveted to part PC.1.
⇒ One `connection_group` between PC.1 & nut.
2. The bolt connects PA.1, PB.1, and PC.1.
The nut is modeled "`fixed_to`".

The discussion showed that this type of modelling is feasible. It can be described using the published standard.

GH will add a note to STEP's recommended practices and also post a comment on issue #106.

The issue will then be closed. // Done 2026-07-27.

I created this slide
after our meeting.
Is it correct, so far?
2026-09-16: Confirmed.

Next meetings — Suggested new appointments

- 2026-10-02, 14:00 CEST — focus group on [#105](#) / [#61](#) (out of series)
 - objective: agree on XML suggestions & examples / draft Word file changes
- 2026-10-07, 15:00 CEST — focus group on [#105](#) / [#61](#)
 - objective: agree on XML suggestions & examples / draft Word file changes

CF sent / will send out invitations.

